

Developer Requirement

GST e-Way Bill API Integration

Objective

Integrate the GST e-Way Bill API into WhizUnik so that users can generate, view, update, cancel, and track e-Way Bills from the existing Sales Invoice and Dispatch workflow.

The e-Way Bill integration must not create duplicate invoices, duplicate inventory movements, or duplicate accounting entries.

1. Core Workflow

Keep the existing sales workflow unchanged.

Sales Invoice Created and Approved

- Generate IRN, if e-invoice applies
 - Generate e-Way Bill, if required
 - Confirm Physical Dispatch
 - Existing inventory debit takes place
 - Customer payment is recorded later, if sale is on credit
- Quotation and Sales Order remain optional.

WhizUnik must support both:

Direct Invoice → IRN → e-Way Bill → Dispatch → Payment
and:

Quotation → Sales Order → Invoice → IRN → e-Way Bill → Dispatch → Payment

2. Important IRN Rule

There are two e-Way Bill generation routes.

Route A — Invoice Requires IRN

For an invoice where e-invoicing is applicable:

Approved Sales Invoice

- Generate e-Invoice / IRN through approved IRP or GSP
- Send transport details with the IRN request
- Receive IRN, QR code, acknowledgement, and e-Way Bill details

Do not use the normal direct e-Way Bill API for B2B or export tax invoices where the supplier is required to generate IRN.

Route B — Invoice Does Not Require IRN

For invoices where IRN is not applicable:

Approved Sales Invoice

- Generate e-Way Bill directly through GST e-Way Bill API
- Receive e-Way Bill number and e-Way Bill date

The application must decide the route based on configurable company, GSTIN, customer, and transaction settings. Do not hard-code tax rules.

3. User Experience

On the existing Sales Invoice or Dispatch screen, show:

Generate e-Way Bill

The button should be available only after the invoice is approved.

The system must automatically populate invoice information and ask the user only for missing transport information.

User Inputs Required

- Transport mode: Road / Rail / Air / Ship
- Transporter name
- Transporter ID or GSTIN / TRANSIN
- Vehicle number, where available
- Vehicle type: Regular / Over Dimension Cargo
- Transport document number and date, where applicable
- Distance
- Dispatch location
- Delivery location

4. Required Data Mapping

GST e-Way Bill API field	WhizUnik source
docNo	Sales Invoice Number
docDate	Sales Invoice Date
docType	Invoice / Credit Note / Debit Note
fromGstin	Company GSTIN / dispatch GSTIN
fromTrdName	Company legal name
fromAddr1, fromPincode,	Warehouse / dispatch address
toGstin	Customer GSTIN
toTrdName	Customer legal name
toAddr1, toPincode, toStateCode	Delivery address
shipToGSTIN, ship-to details	Delivery / ship-to customer details
supplyType, subSupplyType	Transaction configuration
transactionType	Bill-to / ship-to / regular movement configuration
totalValue, totInvValue	Invoice totals

cgstValue,sgstValue,igstValue,cessValue	Invoice tax totals
productName,productDesc	Invoice item / catalogue
hsnCode	Catalogue HSN field
quantity,qtyUnit	Invoice line item
Tax rates	Invoice line item
transporterId,transMode,	Dispatch record
vehicleNo,vehicleType,transDocNo	Dispatch record

The Catalogue must therefore contain valid HSN code and unit-of-quantity details for all products requiring e-Way Bill generation.

5. Direct e-Way Bill API Process

The integration must run from the WhizUnik backend only.

Approved Invoice + Dispatch Information

- Backend validates required data
- Backend authenticates with GST e-Way Bill system
- Backend encrypts and sends Generate e-Way Bill request
- GST system returns success or error
- WhizUnik saves the result against the Invoice and Dispatch record

Authentication Data

Each customer GSTIN must have separate, encrypted credentials:

- Client ID
- Client Secret
- GSTIN
- API username
- API password
- Authentication token
- Session encryption key

Do not expose these credentials in the browser, source code, chat messages, spreadsheets, or client-facing screens.

6. Data to Save After Successful Generation

Save the following fields against the Sales Invoice and Dispatch record:

- e-Way Bill number
- e-Way Bill generation date and time
- e-Way Bill validity date and time

- Transporter name and ID
- Vehicle number
- Vehicle type
- Transport document number and date
- e-Way Bill PDF or printable document
- API request reference
- API response reference
- Status: Generated

Show this information directly on the Invoice and Dispatch page.

7. Status Fields

Add these fields:

Field	Values
E-invoice status	Not Required / Pending / IRN Generated / Failed / Cancelled
E-Way Bill status	Not Required / Pending / Generated / Failed / Cancelled / Expired
Dispatch status	Not Dispatched / Partially Dispatched / Dispatched / Cancelled
API sync status	Pending / Success / Failed / Retry Required

8. Inventory Rule

Generating an IRN or e-Way Bill must not create an inventory movement.

Invoice Created = No additional stock debit

IRN Generated = No stock debit

e-Way Bill Generated = No stock debit

Dispatch Confirmed = Existing stock debit only

The e-Way Bill is a linked statutory compliance record. It is not a sales transaction or stock movement.

9. Duplicate Prevention

Before generating an e-Way Bill, check whether the invoice already has a valid e-Way Bill.

Use a unique internal integration key:

Company GSTIN + Invoice Number + Financial Year + Document Type

Save each API request with:

- source invoice ID;
- source dispatch ID;
- request reference;

- API provider reference;
- request date and time;
- response date and time;
- retry count;
- result status; and
- error code and error message, where applicable.

If a user clicks the button twice, do not create a duplicate request or duplicate e-Way Bill.

10. Error Handling

If the API returns an error, show a clear business-readable message.

Examples:

- Customer GSTIN is invalid
- HSN code is missing or invalid
- Tax values do not match the invoice
- Distance is missing or not accepted
- Vehicle number format is invalid
- Transporter ID is invalid
- e-Way Bill already exists for this invoice
- Invoice date is outside the permitted period
- e-Way Bill must be generated through the IRN route
- API service is unavailable; retry later

The user should be able to correct the relevant information and retry if authorised.

11. Other API Actions Required

Build these actions after basic generation is working:

Action	Purpose
Get e-Way Bill Details	Retrieve and verify e-Way Bill details
Update Part B / Vehicle	Add or change vehicle details
Update Transporter	Change transporter assignment
Cancel e-Way Bill	Cancel where invoice or dispatch is cancelled
Extend Validity	Extend e-Way Bill validity where allowed

Consolidated e-Way Bill	Link multiple invoices moving in one vehicle
-------------------------	--

12. Cancellation and Correction

Before IRN / e-Way Bill Generation

Allow normal invoice cancellation or editing.

After IRN / e-Way Bill Generation

Do not allow silent editing of invoice, tax, customer, quantity, or value fields.

Show a warning:

“This invoice has a statutory document linked to it. Cancel the linked e-Way Bill and/or IRN before cancelling or replacing the invoice.”

Where applicable:

Cancel e-Way Bill

→ Cancel IRN / e-Invoice

→ Cancel or replace Invoice

For corrections, use the appropriate business process:

- cancel and recreate invoice, where permitted;
- issue Credit Note; or
- issue Debit Note.

13. Tally Integration Rule

Do not allow both Tally and WhizUnik to independently generate IRN or e-Way Bill for the same invoice.

Configure one route per client:

Option A:

WhizUnik generates IRN and e-Way Bill

→ Invoice and compliance details sync to Tally

or:

Option B:

Tally / approved accounting provider generates IRN and e-Way Bill

→ IRN and e-Way Bill details sync back to WhizUnik

The source of truth must be agreed before go-live.

14. User Permissions

User	Permission
Salesperson	Create draft invoice
Sales / Operations Manager	Approve invoice and request e-Way Bill

Dispatch / Warehouse User	Add transporter, vehicle, and dispatch details
Finance User	View invoice, IRN, e-Way Bill, payment, and sync status
Admin / Compliance User	Generate, cancel, retry, update, and manage exceptions

Every generation, cancellation, retry, update, and override must be recorded in the audit trail.

15. Acceptance Criteria

The feature is complete when:

1. A user can generate an e-Way Bill from an approved Sales Invoice and Dispatch record.
2. The system chooses the IRN route where e-invoice is applicable.
3. The system uses the direct e-Way Bill API only where permitted.
4. The system saves e-Way Bill number, date, validity, transporter, vehicle, and document details.
5. IRN and e-Way Bill generation do not create duplicate sales, stock, or accounting entries.
6. Inventory is debited only at the existing dispatch-confirmation point.
7. The system prevents duplicate e-Way Bill generation for the same invoice.
8. The user can retrieve, print, update, cancel, or extend e-Way Bill details where authorised.
9. API failures are clearly displayed and safely retried.
10. Tally and WhizUnik cannot generate duplicate statutory documents for the same invoice.
11. All actions are retained in the audit trail.
- 12.